

Simulation and parameters estimation of potassium alum dodecahydrate thermal decomposition

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The present work aims the estimation of the unknown kinetic parameters of a mathematical model previously developed in literature, capable of describing the mass loss of potassium alum dodecahydrate during thermogravimetric analysis, considering the presence or absence of a reducing agent. This work also proposes a thermodynamics assessment of this thermal decomposition of $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, with the intent of investigating the behavior of the molar amounts of $\text{KAl}(\text{SO}_4)_2$, Al_2O_3 and K_2SO_4 as function of temperature and the amount of the reducing agent, simultaneously. The obtained results showed the used model is adequate to represent this reaction system since its simulated results served as good approximations of the experimental TGA data. The model's final predictions of weight percentages for reducing and non-reducing conditions were of 28.16 % and 29.11 %. The calculated activation energies ranged from $1.29 \text{ kJ} \cdot \text{mol}^{-1}$ to $277 \text{ kJ} \cdot \text{mol}^{-1}$, while the Arrhenius constants ranged from 0.199 to $1.33 \cdot 10^{13} \text{ s}^{-1}$. The estimated values of the kinetic parameters were mostly in agreement with the thermodynamic investigations executed regarding this system. . The results obtained by the assessment lead to the conclusion that the reducing agent could be a useful addition within this reaction system, particularly in a full-scale operation, as the decomposition starts and finishes at lower temperatures in its presence.

Keywords: Potassium Alum; Particle Swarm; Kinetic Modeling; Thermal decomposition; Potassium Sulfate

1. Introduction

With an inevitable growth in the world's population and therefore an increase in the world's food demand, the agricultural sector will have to adapt and boost its total production in order to attend this necessity. Within this context, there is also the expectation of a significant expansion in fertilizers' market, due to its capacity to accelerate and spark plants' growth, improving crops' productivities.

Fertilizers are sources of nutrients, chemical compounds or elements that are key to a plant's cycle of life. Amongst the different types of nutrients supplied, potassium is one of the most important, as it exerts an extensive list of crucial functions to a plant's organism wellbeing. The provision of this nutrient to crops is mostly in the form of KCl or K_2SO_4 . Even though the former accounts for the majority share of the total market consumption, the latter holds major importance in specific situations, like in cases where crops are either highly sensitive to chloride anions or are demanding of a robust sulfur supply (Brocchi et al., 2015).

Due to these reasons, there has been an increasing interest in the commercialization and consumption of potassium sulfate fertilizers over the past few years. Currently, there are several processes applied to the large-scale production of this compound such as Mannheim and Hargreaves (Lippman and Comstock, 1943; Mientka et al., 2008). Nevertheless, the demand for efficient potassium sulfate fertilizers at competitive market prices motivates researches and studies focused on developing alternative chemical routes to originate this compound.

One that has been studied recently is the leaching of glauconitic ore with sulfuric acid, followed by the thermal decomposition of potassium alum ($KAl(SO_4)_2$). This mineral is reportedly a good source of potassium as it contains in its composition K_2O ranging from 6 to 14% (Prakash and Verma, 2016). By leaching it with sulfuric acid, there

is the formation of many different compounds, including potassium alum. The thermal decomposition of this double salt is the main reason why this route is attracting plenty of attention, after all, not only it produces potassium sulfate but also aluminum oxide (Al_2O_3) with low concentrations of iron and also no Red Mud formation. This byproduct could have large appliance in the ceramic industry, due to its resistance to corrosion (Brocchi et al., 2015; Habashi, 2005). The formation of two major components with great market potential adds value to this route and also makes a large-scale application of this method of production worth of an investigation.

Brocchi et al (2015) were among the first to propose this route. The authors evaluated the feasibility of this method's application in potassium sulfate production, by analyzing the possibility of precipitating high purity potassium alum after the acid leaching, and by quantifying the yield of this salt's thermal decomposition. They concluded that the application of this route was indeed viable, as they were able to precipitate potassium alum with 99% of purity, and to convert 98% of it into the proposed selectivity (K_2SO_4 and Al_2O_3), while performing the reaction at the temperature of 800°C . The authors also inferred that temperature is vital to optimize the product's conversion, since the yields observed for decompositions occurring beneath 800°C were noticeably lower.

Souza et al. (2018) investigated the thermal behavior of synthetic samples of potassium alum dodecahydrate ($\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$) through thermogravimetric analysis (TGA), under reducing (presence of charcoal) and non-reducing conditions (nitrogen atmosphere). The experiments showed that in non-reducing conditions, the process is conducted by two main transformations (dehydration and the anhydrous double salt decomposition), which occurred in the temperature ranges of 100°C to 250°C and 750°C to 1030°C , respectively. Thermogravimetric analysis of the sample led the authors to

assume that dehydration should take multiple stages to its completion, with two well-defined stages observed. The authors did the x-ray diffraction (XRD) characterization of the intermediate and final products of the thermogravimetric analysis and these characterizations detected the existence of two intermediates ($\text{Al}_2(\text{SO}_4)_3$ and $\text{K}_3\text{H}(\text{SO}_4)_2$) that originate the two main products of interest (K_2SO_4 and Al_2O_3). Combining both data acquired, the authors proposed a 4-stage reaction mechanism concerning the full decomposition of the hydrated salt, with each pair of stages regarding one of the main transformations observed. On the other hand, at reducing conditions, experiments showed the addition of the reducing agent has no effect on the dehydration process. However, it strongly affects the anhydrous salt decomposition, since the reducing agent is responsible of altering the total reaction rate. This has a major impact in the decomposition, as it diminishes the beginning temperature of this process to 580°C , while also modifying the temperature in which the reaction ends.

Envisioning the possibility of a large-scale application of the glauconite route, the reduction of the decomposition temperature is highly desirable financially wise, as it would certainly decrease the operational cost. Thus, it is interest to conduce this process under reducing conditions.. Souza et al.(2018) also suggested that, in the presence of carbon-based materials, the formation of intermediate products would not occur, suggesting a three-stage process, with the decomposition of anhydrous potassium alum happening in just one step.

By acknowledging how the reactions of the process proceed, the next major step to take is the estimation of the unknown kinetic parameters from the reactions that belong to the proposed mechanism (activation energy and pre-exponential constant from Arrhenius equation (Peleg et al., 2012)). With these parameters known, the reaction's participant components behavior can be replicated, allowing the performance of

mathematical simulations for the decomposition considering different temperatures and reagent compositions.

A widely known variety of methods utilized to estimate the kinetic parameters of solids' thermal decompositions are the graphic methods such as Ozawa, KAS and others (Ozawa, 1965; Chee et al., 2016). . The main advantage observed by these techniques is their simplicity and rapid execution; however, they are vastly lacking in precision, especially in complex reaction systems with the involvement of a large number of chemical species (Schwaab et al., 2008). Since the reaction system regarding thermal decomposition of potassium alum dodecahydrate is a complex one, these methods are not suited to estimate the kinetic parameters of this process.

In opposition to the graphic methods, Toniolo et al. (2008) applied an alternative technique of parameter estimation, the same used by Schwaab et al.(2008) . Toniolo et al. (2008) investigated the kinetics of the catalytic combustion of diesel soot with $\text{MoO}_3/\text{Al}_2\text{O}_3$ catalyst. The authors performed the assembled use of high-end computational tools and thermogravimetric analysis data with the intent of calculating the activation energy and the pre-exponential constant of the Arrhenius expression of the combustion reaction rate constant, with precision and reliability far superior than the ones presented by the graphical methods. The authors firstly modeled the molar variation of each specimen in the reaction system within time. With the intent of calculating the kinetic constant each proposed reaction in the model for different values of temperature, they used a reparameterized form of Arrhenius' equation (Schwaab and Pinto, 2007). To adjust the model and consequently estimate the unknown model parameters, they used the particle swarm method to optimize a least squares objective function that takes into account the mass loss calculated by the model and the experimental data provided by the TGA. The results obtained by the authors were extremely satisfying, as they were

coherent chemically speaking, and were able to replicate experimental data with high fidelity.

Other authors adopted this exact approach for parameter estimation. Brandão et al. (2018) used it for estimate the parameters of a model of step-growth polymerization of poly(ethylene-2,5-furandicarboxylate), while Da Ros et al. (2019) applied it in the chemical recycling model of cross-linked poly(methyl methacrylate). Lastly, Vasconcelos et al. (2019) applied the particle swarm optimization to estimate the unknown parameters of the model developed to simulate the solution polymerization of 1,3-butadiene catalyzed by neodymium versatate.

An attempt to do the mathematical modeling of the thermal decomposition of potassium alum dodecahydrate was already made by Nofal et al. (2019). Using Souza's et al. (2018) work as the modeling main foundation, the authors did the thermogravimetric analyses of this compound at heating rates of 10, 15 and 20 K/min, considering reducing and-non reducing conditions (with coke breeze as a reducing agent). They modeled the double salt's mass loss within time and did the adjustment of these models regarding each one of the stages of the full decomposition of the salt, with and without the presence of coke breeze, for all the three heating rates. The developed model showed great accuracy in emulating the TGA data and predicting the system's composition for any given reaction time. On the other hand, the authors just performed simulations with the developed model and a proper estimation of the model unknown parameters was not executed.

The right parameter estimation is essential to have a trustable model. Having good estimations of the values of the parameters is vital when the focus of the work is to shift from laboratory tests to a pilot, and then, to a full-scale industrial production. Therefore, the present study has its objectives related with the proposition of the estimation methods,

previously mentioned and utilized in other literature studies (Schwaab et al., 2008; Toniolo et al., 2008; Brandão et al., 2018; Da Ros et al., 2019; Vasconcelos et al., 2019), in order to determine the values of the unknown kinetic parameters of the mathematical model proposed by Nofal's et al. (2019) for the thermal decomposition of $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ at reducing and non-reducing conditions.

The models were validated and adjusted with the experimental data provided by the TGA experiment performed by Nofal et al. (2019), considering a heating rate of 10 K/min and coke breeze as the reducing agent. The particle swarm optimization method was applied to minimize a least squares function combining experimental data and the simulated results in order to perform this adjustment and provide the best set of the parameters that make the model capable of replicate with the greatest fidelity the mass loss behavior detected by TGA experiments.

Moreover, this study also contemplates a thermodynamic assessment of the reaction system, by means of computational calculations using the software HSC Chemistry, version 9 (Roine, 2002). The thermodynamic investigation aims to provide good predictions of the equilibrium compositions of the major solid species that are participant in the thermal decomposition as a function of temperature and the amount of the reducing agent, in a simultaneous fashion. In addition, the thermodynamics assessment is also vital to confirm the reliability of the kinetics and mechanism propositions regarding the reactions system.

2. Methodology

Experimental Part

The experimental TGA data used in the estimation optimization procedure was extracted from Nofal et al. (2019) work. In their research, samples of 30mg potassium alum dodecahydrate were weighed and then submitted to thermogravimetric analyses.

These experiments were overseen for reducing conditions, with the presence of coke breeze as a reducing agent in a stoichiometric addition, and for non-reducing conditions, in atmospheres of synthetic air and nitrogen. The samples were heated from 25 °C until the temperature of 1000 °C was achieved, and, after hitting this set point, the samples were kept in this same condition for an extended period of an hour.

The experiments were conducted with three different heat rates (10K/min, 15 K/min and 20 K/min), for both conditions. The equipment used in the experiments was a Netzsch thermogravimetric scale model STA 449 F3 Jupiter.

Thermodynamics Assessment

The thermodynamics calculations were performed by using the HSC Chemistry software version 9 (Roine, 2002). A reaction system regarding the presence of all the species and intermediates related to the thermal decomposition of $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ was constructed, and an initial input of 1 kmol of this compound was applied. The equilibrium compositions of the three main solid species present in the system ($\text{KAl}(\text{SO}_4)_2$, K_2SO_4 and Al_2O_3) were then calculated as a function of temperature and the amount of the reducing agent(carbon).

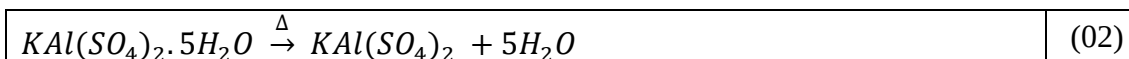
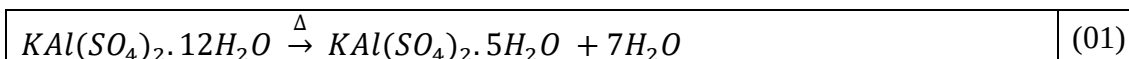
The temperatures utilized in this analysis were between 250 °C and 1000 °C. By defining the lower limit temperature of our analysis at 250 °C, any possible influence that $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ may have in the results was eliminated since, at this temperature, a total conversion of this component into $\text{KAl}(\text{SO}_4)_2$ should be already observed. Thus, this temperature range was set in order to avoid any inconsistencies between the thermodynamics assessment results and the provided TGA data. In extent, the amounts of carbon used in the calculations varied from 0 to 2 kmol since the initial input of the reagent was of 1 kmol. This range of carbon amount of carbon was more than appropriate

for the performed investigations, since it contemplated the effect of carbon at limiting, stoichiometric and excess conditions.

Mathematical Modeling

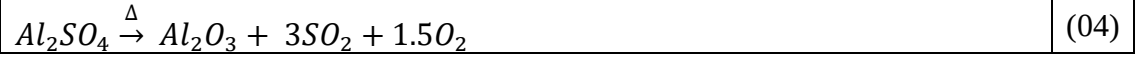
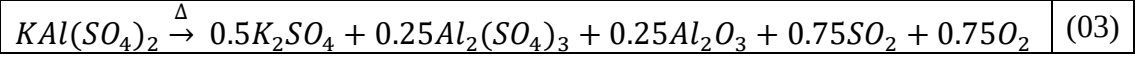
The modeling described here was based on Souza's et al. (2018) proposition of the thermal decomposition's mechanism with and without the presence of the reducing agent, as one of the modeling's main foundations.

For non-reducing conditions, a full decomposition of potassium alum dodecahydrate consisting on 4-steps occurs. The first two steps account for the salt's dehydration, which occurs with an initial loss of seven molecules of water, followed by the final loss of the remaining five molecules of water attached to the partially hydrated potassium alum.



The following stages refer to the anhydrous salt decomposition. In their work, Souza et al.(2018) mentioned the detection of two intermediates ($Al_2(SO_4)_3$ and $K_3H(SO_4)_2$) that precede the formation of the two main products of interest (Al_2O_3 and $K_2(SO_4)$). In their mechanism proposition, however, they only treated the formation of aluminum sulfate as kinetically relevant.

Therefore, the first stage of the anhydrous salt decomposition results in the formation of potassium sulfate and aluminum oxide alongside aluminum sulfate. The final step describes the complete transformation of the intermediate into aluminum oxide. In both steps, there is also the production of O_2 and SO_2 in their gaseous form, which increase the observed mass loss.

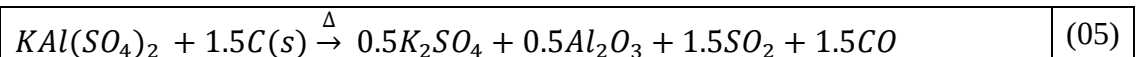


By assuming that all of the reactions listed follow a first order rate and by utilizing the law of mass action, the molar rates are determined, as shown in Table 01. In the expressions constructed, n stands for the mole numbers of the subscript compounds, and k_1 to k_4 stand for the kinetic constants related to Equations 01 to 04 respectively.

Table 01 . Molar rate expressions for non-reducing conditions.

Compound	Molar rate expression
$KAl(SO_4)_2 \cdot 12H_2O$	$-k_1 n_{KAl(SO_4)_2 \cdot 12H_2O}$
$KAl(SO_4)_2 \cdot 5H_2O$	$k_1 n_{KAl(SO_4)_2 \cdot 12H_2O} - k_2 n_{KAl(SO_4)_2 \cdot 5H_2O}$
$KAl(SO_4)_2$	$k_2 n_{KAl(SO_4)_2 \cdot 5H_2O} - k_3 n_{KAl(SO_4)_2}$
H_2O	$7k_1 n_{KAl(SO_4)_2 \cdot 12H_2O} + 5k_2 n_{KAl(SO_4)_2 \cdot 5H_2O}$
K_2SO_4	$0.5k_3 n_{KAl(SO_4)_2}$
$Al_2(SO_4)_3$	$0.25k_3 n_{KAl(SO_4)_2} - k_4 n_{Al_2(SO_4)_3}$
Al_2O_3	$0.25k_3 n_{KAl(SO_4)_2} + k_4 n_{Al_2(SO_4)_3}$
SO_2	$0.75k_3 n_{KAl(SO_4)_2} + 3k_4 n_{Al_2(SO_4)_3}$
O_2	$0.75k_3 n_{KAl(SO_4)_2} + 1.5k_4 n_{Al_2(SO_4)_3}$

For reducing conditions, the mechanism proposed is slightly different. The reducing agent does not interfere in dehydration; hence, the reactions regarding it are similar to Equations 01 and 02. For the anhydrous salt decomposition though, the formation of $Al_2(SO_4)_3$ is suppressed, resulting in the direct production of aluminum oxide, eliminating the need of a fourth reaction to realize the full conversion into the product of interest. The formation of carbon monoxide as a product is also reported, since at the temperatures in which this stage commonly occurs, the reducing agent's carbon tends to combine with oxygen. Therefore, the reaction that dictates anhydrous potassium alum thermal decomposition is the one defined by Equation 05.



Even though the molar rates of species that only participate in the dehydration stages are the same for non-reducing and reducing conditions, the ones related to the anhydrous salt decomposition process suffer major changes. The presence of the reducing agent not only eliminates the need for an additional reaction but also affects the remaining reaction's rate, modifying its expression and kinetic characteristics. It is assumed then, the reaction represented by Equation 05 is an elementary reaction, having a total kinetic order of 2.5, which differs from the first order rate assumed for Equation 03.

Based on what was discussed, the molar rates of all compounds that are originated by the thermal decomposition at reducing conditions are suggested as encountered in Table 02. In these expressions n is the mole number of the subscript compounds, k_{1c} and k_{2c} are the kinetic constants regarding Equations 01 and 02 considering reducing conditions, and k_{3c} is the kinetic constant of Equation 05.

Table 02. Molar rate expressions for reducing conditions.

<i>Compound</i>	<i>Molar rate expression</i>
$KAl(SO_4)_2 \cdot 12H_2O$	$-k_{1c}n_{KAl(SO_4)_2 \cdot 12H_2O}$
$KAl(SO_4)_2 \cdot 5H_2O$	$k_{1c}n_{KAl(SO_4)_2 \cdot 12H_2O} - k_{2c}n_{KAl(SO_4)_2 \cdot 5H_2O}$
$KAl(SO_4)_2$	$k_{2c}n_{KAl(SO_4)_2 \cdot 5H_2O} - k_{3c}n_{KAl(SO_4)_2}n_C^{1.5}$
H_2O	$7k_{1c}n_{KAl(SO_4)_2 \cdot 12H_2O} + 5k_{2c}n_{KAl(SO_4)_2 \cdot 5H_2O}$
K_2SO_4	$0.5k_{3c}n_{KAl(SO_4)_2}n_C^{1.5}$
$C(s)$	$-1.5k_{3c}n_{KAl(SO_4)_2}n_C^{1.5}$
Al_2O_3	$0.5k_{3c}n_{KAl(SO_4)_2}n_C^{1.5}$
SO_2	$1.5k_{3c}n_{KAl(SO_4)_2}n_C^{1.5}$
CO	$1.5k_{3c}n_{KAl(SO_4)_2}n_C^{1.5}$

The modeling of the molar balance equations is the key in order to obtain the instantaneous composition of any reaction compound for a given reaction time. To perform these calculations however, it is crucial to attest the values of the different kinetic

constants that affect the molar rates of all species that participate in the decomposition process.

All described reactions, have their own kinetic parameter that depends on the medium temperature. To achieve the values of these constants for different temperatures, and consequently different reaction times, a reparameterized form of the Arrhenius equation was used. In Equation 06, i varies from 1 to 4 (being related to the constants k_1 to k_4) and $1c$ to $3c$ (regarding the proposed kinetic constants at reducing conditions, k_{1c} to k_{3c}).

$k_i = \exp(A + 10^B (\frac{T - T_{ref}}{T}))$	(06)
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One of the differences that can be noticed between the reparameterized expression and the original Arrhenius equation is the presence of A and B . These two parameters are the ones that are estimated, and the values of the activation energy (E_a) and the Arrhenius pre-exponential constant (k_0) can be recovered from A and B using Equations 07 and 08. In these expressions, R is the gas constant.

$A = \ln k_0 - \frac{E_a}{RT_{Ref}}$	(07)
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$B = \log(\frac{E_a}{RT_{Ref}})$	(08)
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The other perceived alteration within the reparameterized form of Arrhenius equation is the presence of T_{Ref} , which stands for reference temperature, and varies depending on the reaction. This temperature is extremely important, since its addition to the expression diminishes the correlation between A and B , minimizing the error of the final estimations (Toniolo et al., 2008). The values of the utilized reference temperatures are usually a suitable average for the evaluated experimental data (Schwaab et al. 2008) and these are displayed in Table 03.

Table 03. Reference temperatures.

<i>Kinetic constant</i>	<i>T_{ref}(°C)</i>
k₁	155
k₂	220
k₃	800
k₄	800
k_{1c}	155
k_{2c}	220
k_{3c}	748

It is valid to notice the heating rate used in TGA is omitted in these expressions. This is due to the fact the actual measured values of temperature for each interval of time are utilized instead (Conesa et al., 2001).

In the parameter estimation procedure, the models are adjusted by testing different values for the kinetic parameters, so they can replicate with maximum precision the experimental results, which, in this case, are provided by the TGA. The equations presented previous only determine the composition and the mole number of each system's component during time. Thus, in order to perform a direct comparison with the TGA results, the models' simulated compositions were converted into data regarding mass loss. This is done by using Equation 09. This expression is the ratio of the total weight of all the solid components for a reaction time(t) $\neq 0$, with the initial measured weight of $KAl(SO_4)_2 \cdot 12H_2O$.

$mass\ loss_{\%} = \frac{total\ solids\ weight_t}{total\ solids\ weight_{t=0}} \cdot 100$	(09)
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Model Simulation and Parameters Estimation Procedure

The ordinary differential equations presented in the previous paragraphs were solved numerically by using the Runge-Kutta method of 4th order programmed in MATLAB code. The parameters estimation was performed by the usage of the particle swarm optimization method with the intent on minimizing an objective function consisting on the sum of two least squares functions. The purpose was to find amongst

several different combinations of parameters one that provides a value of F_{obj} closer to zero.

The two least squares functions compare both experimental data and simulated data provided by the modeling. One of the functions correlates experimental TGA data with simulated TGA data, while the other contemplates a comparison between simulated and experimental differential thermogravimetry (DTG) data. MATLAB function “spaps” was utilized in order to smoothen the DTG data and eliminate noisy values of differential weight loss. The usage of smoothed DTG results in the objective function was to ensure that the model’s simulations after the parameters estimations would replicate mass loss behavior with maximum precision. The objective function is represented by Equation 10. In this expression, F_{obj} is the objective function, n stands for the number of measurements regarding mass loss, y_i^{exp} is experimental data for mass loss, while y_i^{model} is the simulated mass loss provided by the model and, finally, both dy_i^{exp} and dy_i^{model} are related to the derived mass loss measured and predicted, respectively. The second term of Equation 10 was multiplied by the factor 2 so this term would have a bigger importance in the final F_{obj} value.

$F_{obj} = \sum_i^N (y_i^{exp} - y_i^{model})^2 + 2 * (dy_i^{exp} - dy_i^{model})^2)$	(10)
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The particle swarm optimization was implemented in MATLAB code. In the optimization calculations, three hundred particles were used and a maximum number of three hundred iterations was delimited, as well as the maximum number of twenty stall iterations. Results were obtained whenever the number of iterations achieved three hundred or there was no significant change whatsoever in the minimum estimated value of the F_{obj} over the last twenty iterations run. After the end of the estimations, the values

of A and B are obtained, and, with them, E_a and k_0 can be calculated by the appliance of Equations 07 and 08.

3. Results and Discussion

Thermodynamics Assessment

The thermodynamics assessment investigates the behavior of the equilibrium compositions of $KAl(SO_4)_2$, K_2SO_4 and Al_2O_3 as described in the methodology section. In this context, Figure 01 depicts the equilibrium composition percentage of $KAl(SO_4)_2$ within temperatures in the range of 250°C to 1000°C and the amount of carbon varying from 0 to 2kmol.

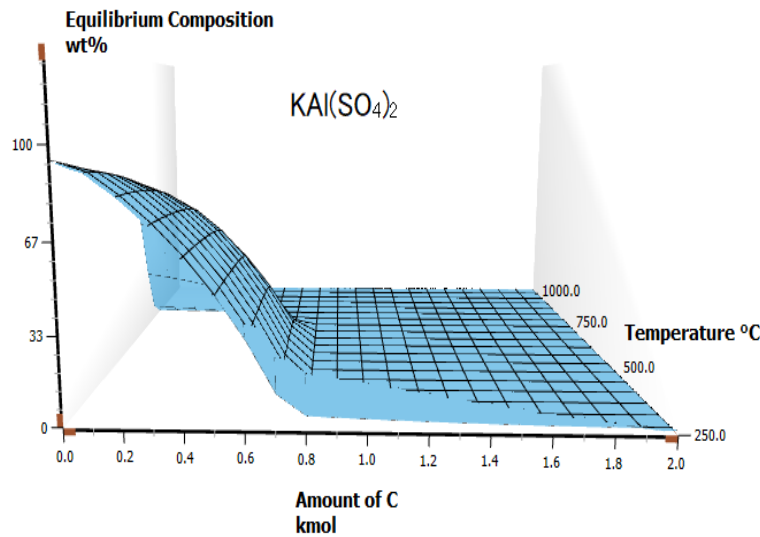


Figure 01 Equilibrium composition of $KAl(SO_4)_2$ as a function of temperature and the amount of Carbon

These percentages only account the weight of the solid species formed in the decomposition process.

At the lower limit conditions, in which the amount of carbon is 0kmol and the temperature is equal to 250°C, it is observed that $KAl(SO_4)_2$ stands as the major solid specie at the system. Considering that the initial input of the assessment was of 1kmol of

$\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, it is clear that this reagent had already been totally converted into $\text{KAl}(\text{SO}_4)_2$ at this point.

With the amount of carbon fixed in 0 kmol, the effect that temperature has in the anhydrous salt decomposition is investigated. For temperatures beneath 500°C it is noticeable that $\text{KAl}(\text{SO}_4)_2$ presents a certain stability, since its equilibrium composition remains practically the same as the one observed at 250°C . When the temperature surpasses 500°C , the equilibrium composition slowly begins to decrease until it hits 750°C , where it starts to fall considerably, rapidly diminishing its values before it reaches 0% at temperatures in between the range of 750°C and 1000°C . This implies that the decomposition of $\text{KAl}(\text{SO}_4)_2$ has a proper beginning near 750°C and ends up consuming all of its reagent at temperatures approximate to 800 and 1000°C .

By doing the exact opposite, and fixing the temperature in 250°C , the effect of the amount of the reducing agent in the decomposition is analyzed. Its addition to the system has a major and immediate impact in the equilibrium composition of $\text{KAl}(\text{SO}_4)_2$. Even for low molar amounts of carbon, in which the reducing agent acts as the decomposition's limiting reagent, it is well observed that the equilibrium composition of potassium suffers a considerable decrease when compared to the calculated value with 0 kmol of the reducing agent. By increasing the amount of carbon, the equilibrium composition continues to plunge, until it hits 0% for an amount of carbon of approximately 0.8 kmol.

The amount of carbon has similar effects at the decomposition process for almost any temperature conditions. The only detected exceptions are for ones ranging from 800 to 1000°C . At these, the full conversion of $\text{KAl}(\text{SO}_4)_2$ is observed even for an amount of 0 kmol of the reducing agent. This implies that the addition of carbon in processes occurring at these temperatures have no effect at the final equilibrium composition.

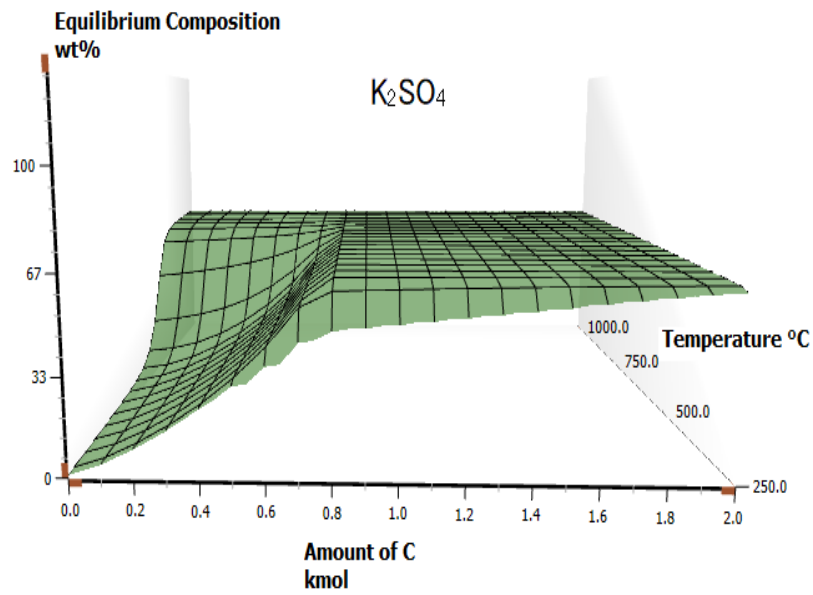


Figure 02 Equilibrium composition of K_2SO_4 as a function of temperature and the amount of Carbon

Figures 02 and 03 provide a complementary picture in the shift in the equilibrium composition presenting K_2SO_4 and Al_2O_3 formation respectively, for same temperature ranges and molar amounts of carbon depicted in Figure 01.

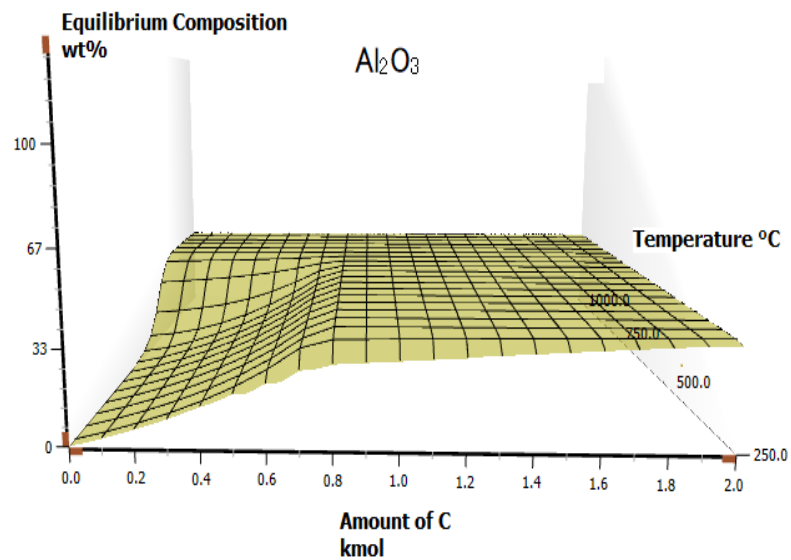


Figure 02 Equilibrium composition of Al_2O_3 as a function of temperature and the amount of Carbon

At the lower limit conditions, it is observed that both compounds have not had its production started yet, with both equilibrium compositions matching 0%.

Since K_2SO_4 and Al_2O_3 are products of the decomposition and share an equal stoichiometry, their equilibrium compositions have an identical behavior within variations in temperature and the amount of carbon. The main difference observed is in the percentage achieved by each component in the equilibrium composition. This is easily explained by the different molar masses that each compound present, and, due to K_2SO_4 having a higher molar mass, this compound accounts as the major specie in the total solids composition.

The depicted behavior for K_2SO_4 and Al_2O_3 is the exact opposite of the one observed for $KAl(SO_4)_2$, which is expected since the latter is the reagent of the process and the former are the products. Thus, the formation of these compounds is only noticed for temperatures above $750^{\circ}C$, and the full conversion of $KAl(SO_4)_2$ into K_2SO_4 and Al_2O_3 , at temperatures close to $800^{\circ}C$ and $1000^{\circ}C$, at least considering an amount of carbon of 0kmol. By taking a good look at both graphs, it can also be inferred that at the end of the decomposition, K_2SO_4 and Al_2O_3 are the only solid species present in the reaction medium. By summing the values of the final equilibrium compositions of both compounds, the value of 100% is achieved, confirming, that these species are the only solids left in the reaction medium.

It crucial to address that even though the envisioned mechanism for the decomposition of $KAl(SO_4)_2$ proposes the production of $Al_2(SO_4)_3$ as intermediate prior to the formation of Al_2O_3 in non-reducing conditions, while analyzing Figure 02 and Figure 03, the formation of $Al_2(SO_4)_3$, wasn't detected. When comparing the two graphs, it was expected expected that at $750^{\circ}C$, temperature in which the decomposition properly begins, the inclinations presented by both diagrams from this point until $1000^{\circ}C$ would

present different behaviors due to the formation of $\text{Al}_2(\text{SO}_4)_3$. Still, despite the inclination shown in Figure 02 being much steeper than the one in Figure 03, this is due to the molar mass differences between both compounds. It is concluded can conclude then, that the formation of $\text{Al}_2(\text{SO}_4)_3$ is possibly reduced in the process, with small only small quantities of this intermediate being produced and quickly being consumed.

As it was noticed for the decomposition of $\text{KAl}(\text{SO}_4)_2$, the amount of carbon has a drastic effect on the formation of Al_2O_3 and K_2SO_4 . The addition of the reducing agent increases considerably the equilibrium compositions of these components, at any temperature below 750°C . Above this mark, the equilibrium composition has the same behavior observed if as one considering an amount of carbon of 0kmol .

A superficial analysis of what was depicted in Figures 01, 02 and 03, could lead to the conclusion that by performing the thermal decomposition with amounts of carbon over 1kmol of carbon and the temperature of 250°C would provide a satisfying production of potassium sulfate and aluminum oxide. It is crucial to remind though, that the thermodynamics assessment only provides data of the calculated equilibrium compositions of each analyzed compound, with the kinetics regarding the thermal decomposition not being contemplated during this process. Even though the results show that for an amount of carbon of 1kmol and a temperature of 250°C , all of $\text{KAl}(\text{SO}_4)_2$ can be converted into Al_2O_3 and K_2SO_4 , it is highly unlikely that the process at these conditions would occur with a productivity that would allow a large scale of production of these products of interest. In order to verify the necessary conditions that, when applied, provide a good conversion of potassium alum dodecahydrate into potassium sulfate and aluminum oxide, the thermogravimetric analysis is needed.

TGA and DTG experimental data

The potassium alum dodecahydrate weight loss and its derivative reported by Nofal et al. (2019) for a heating rate of $10\text{ K}\cdot\text{min}^{-1}$ at reducing and non-reducing conditions is presented at Figure 04, with NRC standing for non-reducing conditions and RC standing for reducing conditions. In this Figure, two y axes are depicted, the left axis is related to the weight percentages measured within the advance of temperature at the TGA experiment, while right axes corresponds to the differential thermogravimetry measurements. From this point on, all the Figures present in this work that depict both TGA and DTG data, will have these y axes.

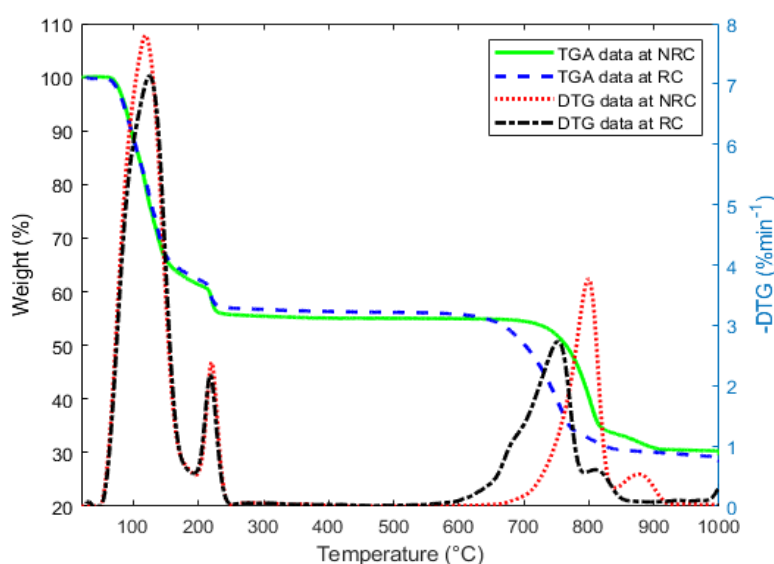


Figure 04. Experimental data obtained at heating rate of $10\text{ K}\cdot\text{min}^{-1}$ extracted from Nofal et al. (2019).

As one can view in Figure 04, both TGA curves present two regions in which huge weight losses are detected. We can infer the first region observed is directly related to the dehydration process of the thermal decomposition of potassium alum dodecahydrate, while the second corresponds to the anhydrous salt decomposition. This can be assumed by correlating the temperatures in which both transformations occur and the temperatures that the weight losses are accounted. The first region is the same for both TGA curves, it begins at a temperature next to $50\text{ }^{\circ}\text{C}$, in which a major plunge in the weight percentages

is easily perceived. The mass loss continues to increase with the temperature rise until the measured weight percentages reach nearly 60 %. When it seems that these weight percentages will stabilize, an inflexion point in the curves is observed, and the weight loss continue to occur until a temperature near 250 °C is reached. From this point on, the curves regain their stability, and the weight percentage suffers no variations for an extended range of temperature values, sealing then, the end of the first region.

This scenario is in accordance with what was expected since the addition of the reducing agent has no effect whatsoever in the dehydration transformation. This becomes very apparent when evaluating the DTG curves also presented in Figure 04. At the dehydration region, no major differences are detected within data regarding weight loss considering both reducing and non-reducing conditions. The only difference between both DTG curves lies in the values achieved by the first differential weight percentage peak of each curve, whereas at non-reducing conditions, this peak is a bit higher than the one observed at reducing conditions, however it is a slight difference. Still inside the first region, two major peaks are recognized within both DTG curves. The first peak is encountered at the temperature of 118 °C, while the other is at 220 °C. This information confirms what Souza et al. (2018) concluded in their work, regarding how the dehydration step proceeds, as the existence of the two major peaks in both DTG curves suggests the dehydration mechanism has at least two reactions. Moreover, as aforementioned, the addition of the reducing agent has no effect whatsoever in the dehydration step.

The second region that can be identified in Figure 04 starts a little beneath 600 °C for reducing condition and 680 °C for non-reducing condition. It is clear the effect the reducing agent has on the dislocation of the TGA curves and of the DTG peaks. After the beginning of the anhydrous salt decomposition process, the behavior of both TGA curves

follows a similar pattern. At reducing condition, the weight percentage ceases to decrease at a temperature close to 850 °C, resulting in 28.18 % of the original sample weight, which is inferior to the 29.66 % achieved at non-reducing condition. The analysis of these graphs indicates the reducing agent mostly affects the temperatures necessary in order achieve a full conversion into the major products of interest. Observing the DTG curves at the second region, it is possible to clearly identify two differential peaks for the non-reducing condition. Due to the influence of the reducing agent on the temperature in which the anhydrous salt decomposition occurs, the two peaks for the reducing condition also can be observed but at lower temperatures when comparing to the ones observed at non-reducing condition, which agrees with what was discussed before. However, the DTG peak values at reducing condition also appear to be smaller and the addition of the reducing agent seems to suppress the second differential weight loss peak which is mixed with the first DTG peak of the second region. These results match with what was proposed by Souza et al. (2018) , that the mechanism regarding the anhydrous salt decomposition only takes a single stage to its full completion. Still, it is possible this transformation has other stages regarding its mechanisms.

Non-reducing condition

For the non-reducing condition, two reaction channels were considered for the alum dehydration step and also two reaction channels were considered for the anhydrous salt decomposition step as shown in Equations 01 to 04. In the total, eight model parameters were estimated, two for each kinetic reaction rate constant. With the parameters estimated, the model was run to predict the weight loss of the thermal decomposition and its derivative of potassium alum dodecahydrate as a function of temperature.

Figure 05 draws a parallel between the TGA and DTG data primarily displayed in Figure 04, with the simulated results concerning the mathematical model developed to represent the thermal decomposition of alum dodecahydrate at non-reducing condition.

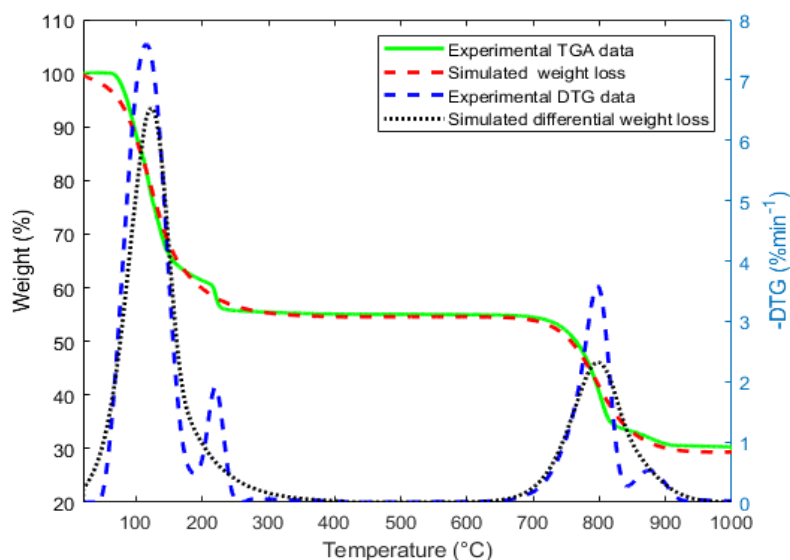


Figure 05 Comparison between experimental and simulated data at non-reducing condition.

An initial analysis of the simulated data into Figure 05 shows the weight loss related to the dehydration process starts at temperatures below 20 °C. This inference about the beginning temperature of this transformation is discrepant with what was attested regarding the TGA experiment since, as previously discussed, the first sign of weight loss in the thermogravimetric analysis was at 50 °C.

After this mass percentage decay happening earlier than expected, and at a temperature above 100 °C, the simulated weight percentages start matching with good precision the results provided by the experimental thermogravimetric analysis. Figure 05 indicates that at this temperature, the two weight loss curves overlap each other and, even with the temperature increase, both curves continue to display analogous results.

Additionally, Figure 05 also demonstrates the behavior observed for the simulated differential weight percentage curve followed a similar pattern to the behavior expressed

by the experimental DTG curve within temperatures between 50 °C and 120 °C, with both curves presenting differential weight percentages peaks at comparable temperatures.

With the advance in the temperature values, the model's fits for the weight percentages as function of temperature continue to provide results very close to the ones provided by the TGA until the end of the dehydration transformation. As a matter of fact, the highest difference observed by TGA data and the model's fits for weight percentages was of 3.76 %, which was observed at the beginning temperatures of the process. This difference is due to the bad prediction of the model regarding the thermal decomposition beginning temperature. Even so, it is safe to assume the model's fits are able to simulate properly the experimental data measure by the TGA.

Nevertheless, even if the model's fits serve as good approximations of the weight percentages measured during the thermogravimetric experiment, this does not mean the model is able to describe faithfully the weight loss behavior observed within the analysis.

At a temperature near 220 °C, the experimental TGA curve presents an inflexion point that consequently changes the curve's behavior and causes the weight loss to increase. This inflexion point is not encountered in the simulated results' curve. The comparison between the two DTG curves at this temperature range complements this discussion. After the first differential peak, the simulated results' differential weight percentage curve initiates to fall and continues its descending until its value reaches zero at a temperature above 300 °C. In opposition, the differential curve regarding the experimental data decreases a little, but then, at the temperature 200 °C, the curve returns its ascendance until it achieves a second peak at 220 °C. After that, it falls rapidly until it reaches zero at 250 °C. The presence of this second peak explains the existence of the inflexion point in the experimental TGA curve while the absence of this peak in the

simulated DTG curve justifies why an inflexion point is not in the weight loss curve for the simulated results.

The discrepancy in both DTG curves also clarifies the considerable difference regarding the simulated results' prediction for the temperature in which the dehydration terminates and the actual temperature that marks the end of this transformation. The model's fits propose the dehydration would only be concluded at 350 °C. This does not agree with what was inferred by analyzing the experimental data. As previously discussed, the experimental data suggests the dehydration would be finished at a temperature somewhere approximate to 250 °C. Hence, the model's simulated results fail to predict accurately the final temperature of the dehydration and to describe the weight loss regarding this region.

At the end of this transformation, the calculated weight percentage by the model's simulations was of 54.56%, a value close to the 55 % obtained by thermogravimetric analysis. Posterior to the dehydration, both the TGA profile and the model's fits continued unaltered until the beginning of the anhydrous salt decomposition.

Both experimental data and model's simulations point to 680 °C as the temperature that ignites the start of the decomposition process, as both DTG curves reinitiate to ascend at similar fashion. Hence, the model's prediction of the beginning temperature of the decomposition proves itself to be accurate. It is easily noticeable that both TGA curves for the model's fits and experimental data present very similar weight percentage values within the temperature raise, with both curves overlapping each other constantly. The major deviance between the experimental data and the model's fits observed at this region was of 2.85 %, at the temperature of 815 °C. This difference is minor, and only complements to the conclusion that the model is also able to simulate the

weight percentages during the advance of temperature at the decomposition of the anhydrous salt region.

Despite this, the model's fits differential weight loss profile do not match the experimental data's weight loss behavior. This is very apparent by comparing the two DTG curves. Even though they both display a similar behavior in the initial stages of the weight loss, after 720 °C, the two DTG profiles start to differ considerably. The two curves keep on raising their differential weight percentages values with the increase of temperature, until both reach a differential weight loss peak at 800 °C. From this point on, the model's fits DTG curve decreases slowly until it reaches zero, while the experimental data curve presents a similar behavior to what was observed at the dehydration process. The behavior of these two differential curves help us understand the reason why at 815 °C the biggest difference between the predicted and experimental percentages is detected. After the peak at 800 °C, the model's DTG curve starts to decrease slowly and the experimental DTG curve plunges rapidly. At 815°C, while the simulated DTG curve presents a differential weight percentage value next to the peak at 800 °C, the experimental DTG curve ceases to decrease and initiates to raise its values again.

The model predicts the end of the anhydrous decomposition occurs a little above 920 °C. This temperature is very close to the attested at the thermogravimetric analysis, which entails in another good prediction of the model, regarding the decomposition region.

If one take a deeper glance at the DTG curves' profile at the decomposition region, it is possible to see the value of the differential weight percentage peak in the model's fits curve is approximate to an average of the two peaks' values of the experimental data curve. If conserved a hypothesis where the experimental DTG curve

was condensed into one with a single differential weight percentage peak, the final resulting curve would share a high resemblance with the profile predicted by the model, especially since both curves initiate and terminate at equivalent temperatures for the decomposition transformation. This helps to understand the reason why the model succeeds in simulating the experimental weight percentages attested in thermogravimetric analysis but fails in describing the mechanism that dictates this weight loss.

At the end, the final weight percentage estimated by the model was 29.11 %, which is a good prediction of the 29.66 % measured at end of the thermogravimetric analysis. Overall, the analysis of the second region serves to reinforce what was discussed before. Despite some of the mentioned discrepancies between the models simulated results and the experimental data, it can be affirmed that the model's fits were able to provide good predictions of the experimental data at the decomposition region

Table 04 shows the estimated values of each of the unknown model parameter of the developed model.

Table 04. Estimation of the model's unknown parameters at non-reducing conditions.

Equations	$k_0(s^{-1})$	$E_a(KJ.mol^{-1})$
1	$1.44 \cdot 10^{05}$	40.97
2	0.199	1.297
3	$1.330 \cdot 10^{13}$	277.0
4	$3.59 \cdot 10^3$	89.23

Each of the four pairs of parameters is related to a reaction step proposed by the utilized mathematical model in order to describe the thermal decomposition of potassium alum dodecahydrate. The first step, represented by Equation 01, accounted the loss of seven molecules of water by the hydrated double salt. The Arrhenius constant and the activation energy of this reaction are $1.445 \cdot 10^5 s^{-1}$ and $40.97 \text{ kJ} \cdot \text{mol}^{-1}$, respectively.

As previously discussed, although the model predicted a beginning in the weight loss of the thermal decomposition earlier than supposed to, it was still able to emulate the behavior of weight loss and differential weight loss with a certain fidelity until the temperature of 120 °C. At this temperature, a differential weight loss peak was observed both in the simulated results' DTG curve, and in the experimental DTG data. The presence of a differential mass loss peak in the DTG curves is usually related to reactions that cause mass loss in the thermal decomposition process. Since the simulated results replicated DTG data with certain fidelity, it was assumed the obtained kinetic parameters and the model's first step proposition provided good predictions on this first water loss kinetics and mechanism. Hence, they are appropriate to describe this observed mass loss within the thermal decomposition at this temperature range.

The same cannot be inferred for the second proposed step in the thermal decomposition. The mathematical model proposes a second step accounting the loss of five molecules of water to finalize the dehydration process. As shown in Table 04, the Arrhenius pre-exponential constant and the activation energy of this reaction are equal of 0.199 s^{-1} and $1297 \text{ J}\cdot\text{mol}^{-1}$.

As seen in this previous topic, the model did not succeed in replicating the behavior of mass loss after 120°C until the end of dehydration. The values estimated for the kinetic parameters is helpful to understand the reason why the model did not succeed in describing appropriately the experimental data. The value of 0.199 s^{-1} obtained for the Arrhenius pre-exponential constant explains why the model predicted that the dehydration process would only terminate at temperatures above 300 °C. A low value in the Arrhenius pre-exponential constant, which is what is observed in this case, entails in a diminished reaction rate. Thus, for this reaction to undergo a full completion, it would need to be performed at high temperatures. This justifies why the model predicts that the

dehydration transformation would only terminate at a much higher temperature than it was depicted by experimental data.

Moreover, the value obtained for the activation energy also explains why there is not a second mass loss peak in the simulated results' DTG curve. A reaction does not occur if the system does not receive an energy that equals or surpasses the reaction's activation energy. The estimated activation energy for the reaction regarding Equation 02 was lower than the one observed for Equation 01, which means that Equation 02's reaction begins at lower temperatures than Equation 01. Since the second step of the dehydration can only begin after the first one starts, due to the initial absence of $\text{KAl}(\text{SO}_4)_2 \cdot 5\text{H}_2\text{O}$ in the system, it can be concluded the model's simulations predict that the two dehydration steps are simultaneous, with the second one beginning with the formation of the first molecule of $\text{KAl}(\text{SO}_4)_2 \cdot 5\text{H}_2\text{O}$. With these two proposed reactions occurring simultaneously, the absence of the second differential weight loss peak is noticed in the simulated results curve. At the end, the model could not describe with much precision the loss of the remaining molecules of water in the molecule of the hydrated double salt.

The final two steps in the thermal decomposition's mechanism are related to the decomposition of the anhydrous salt and are represented by Equation 03 and 04. Equation 03 is related to the salt's conversion into aluminum and potassium sulfate. The Arrhenius pre-exponential constant and the activation energy related to the third step correspond of $1.330 \cdot 10^{13} \text{s}^{-1}$ and $277.0 \text{kJ} \cdot \text{mol}^{-1}$, respectively. Equation 04 depicts the transformation of aluminum sulfate into aluminum oxide, with the values of the Arrhenius pre-exponential constant and activation energy equal to $3.593 \cdot 10^3 \text{s}^{-1}$ and $89.23 \text{kJ} \cdot \text{mol}^{-1}$, respectively

The lower value of the activation energy of Equation 04 when comparing to Equation 03 portrays a scenario similar to the dehydration region. Since Equation 04

contemplates the transformation of aluminum sulfate into aluminum oxide, and the former is not produced until the reaction depicted in Equation 03 begins, it can be deduced the model predicts that both reactions are simultaneous. The absence of a second peak in the simulated results DTG curve is also explained by the lower activation energy of Equation 04. It is interesting to perceive that, meanwhile the activation energies for the two reactions present values not so different from each the other, the Arrhenius constants and, consequently, the estimated reaction's rate differ immensely. A further look into the predicted composition of each component in the system will give a better insight over this subject

Due to what was already discussed, the estimated parameters for Equation 03 and Equation 04 prove that the model is suitable to simulate the weight percentages of the thermal decomposition of potassium alum, within the advance of temperature.

With the kinetic parameters estimated, the composition of each component participant in the reactions is simulated within the advance of temperature. The input composition considered in the simulations was of 1kmol of potassium alum dodecahydrate. Figure 06 displays the total solids composition percentage as a function of temperature simulated by the developed model.

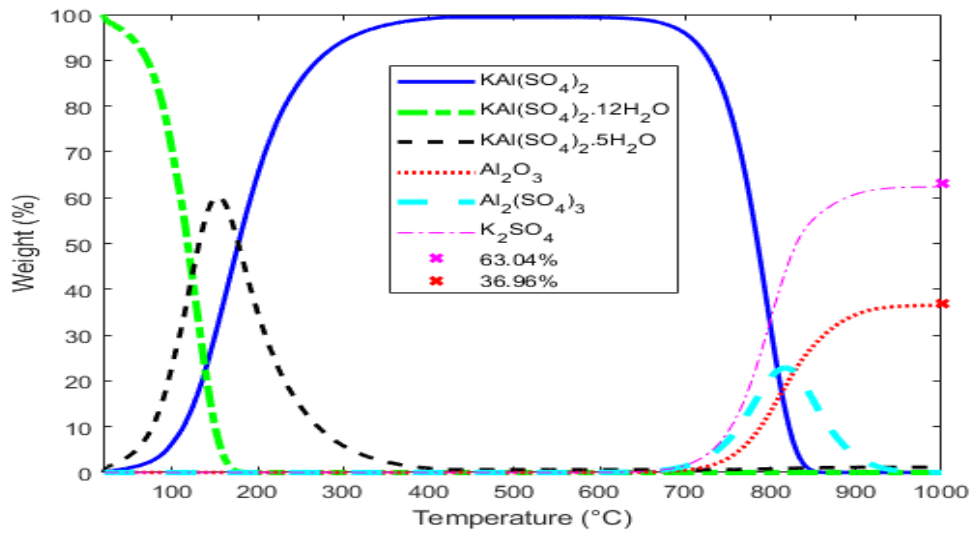


Figure 06 Model's predicted solids composition

The profile of the solid species displayed in Figure 06 re-presents what was portrayed and discussed in previous topics. The model predicted an early beginning and a late ending for the dehydration process. This is well apparent in Figure 06 by judging the profiles of potassium alum dodecahydrate and the partially hydrated alum. The consumption of the former starts right at the first temperatures of the graph attesting the early predicted of the model. Meanwhile, the latter is only fully transformed into the anhydrous salt at temperatures after 300°C, confirming the model's late predictions. At the decomposition region, it is noticed that the anhydrous potassium alum begins its transformation into the products of interest 680°C. The decomposition of this salt is completed at 850°C, with its total composition percentage reaching 0%. Yet, at this temperature, there is still a considerable amount of aluminum sulfate in the reaction system. This is due to the estimated pre-exponential constant that is correlated to Equation 04, which is considerably lower than the one calculated for Equation 03. The difference is the calculated values for these constants explains why the reaction represented by Equation 04 terminates at a later temperature. More information in the predicted compositions of each component is displayed in Appendix I.

One thing worth noting about this Figure is the percentage achieved by K_2SO_4 and Al_2O_3 at the end of the thermal decomposition. In Souza's et al. (2018) work, the authors affirmed that at the end of the thermogravimetric analysis, the final mixture should consist of pure K_2SO_4 and Al_2O_3 with mass fractions of 63.04 % and 36.96 %. At the end of the simulated thermal decomposition, the mass fractions of these components were of 62.36 % and 36.49 %. These values strongly agree with what was proposed by the authors and are sign of a great prediction by the model.

Reducing conditions

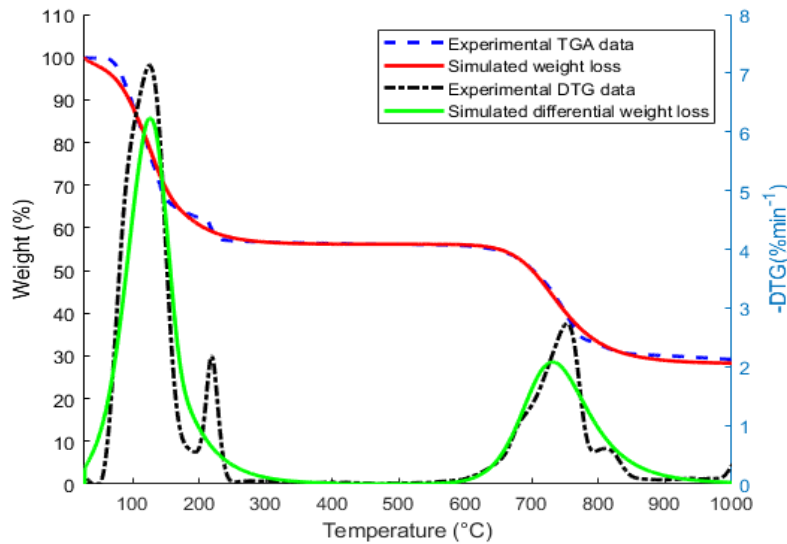


Figure 07 Comparison between experimental data and the simulated results at reducing conditions

Figure 07 compares both the weight loss and the differential weight loss predicted by the model with the experimental TGA and DTG data at reducing conditions. A stoichiometric input of 1kmol of potassium alum dodecahydrate and 1.5 kmol of Carbon is considered.

It is perceived by analyzing this Figure that the model's predicted behavior at the dehydration region is exactly the same one observed at non-reducing conditions. As remarked in previous topics, the dehydration process is not affected by the addition of the

reducing agent. This was bolstered by the evaluation of Figure 04, where it was attested that, at this region, both curves regarding reducing and non-reducing conditions shared a similar profile. In addition to that, the proposed mechanism for the dehydration at reducing condition, is equal to the one proposed for non-reducing conditions.

Therefore, even with us readjusting the model for the new TGA and DTG data considering the presence of coke breeze, recalculating the model's kinetic parameters and the model's predictions for weight loss, the simulated result for the dehydration region and its relation to experimental data did not change. Thence, all that was commented about the model's qualities and flaws within this region also apply for reducing conditions.

At the end of the dehydration region, the weight percentage predicted by model was of 56.18%. In opposition to the 56.26% measured by experimental data. With a difference of less than 0.1%, it is safe to assume that model's prediction succeeds in replicating the experimental data. After the dehydration, the profiles remain stagnant until the beginning of the decomposition.

Figure 07 shows that, within the temperatures of 500°C and until a little above 700°, the profiles of the differential weight percentages curves of both model's fits and the experimental data are extremely identical. As a result of this, the model's prediction of the beginning temperature of the decomposition at reducing conditions matches greatly with the actual temperature that marks the start of this transformation.

It is apparent that the model's fits provide a very good estimation of the thermogravimetric experiment realized at reducing conditions. By analyzing the TGA curves along the extent of the decomposition, it is simple to note that the two curves display very alike weight percentage values and tend overlap each other frequently.

However, the weight loss profiles demonstrated by the model's fits and the experimental data continue to deviate, even at reducing conditions. After 700°C, the profiles of the two DTG curves start to distinct themselves. The experimental DTG curve continues to raise alongside temperature until 750°C. After that, the curve faces a decline until 800°C where it restarts to rise until it reaches a peak at 810°C. The behavior observed by the model's fits DTG curve do not agree with the experimental curve. A little before 750°C, the simulated curve reaches a differential weight percentage peak, and not long after achieving this value, it starts to slowly decrease until it reaches zero, defining the end of the decomposition.

The model predicts that a temperature above 900°C would mark the end of the transformation. This prediction was a little late when compared to actual temperature that determines the ending of decomposition, which is of 850°C. As well as it was observed in non-reducing conditions, the differential peak observed for the model's fits is very close to the average of the two differential peaks of the experimental curve. Therefore, the inference done regarding the relation between the two DTG profiles at the decomposition region in the absence of coke breeze, maintains itself relevant in this topic. It is crucial to infer as well, that the model's proposed mechanism for the decomposition at reducing conditions, contemplates a single step in order to its full completion. We notice in the DTG experimental curve the presence of two weight loss peaks, even if the second is a bit suppressed. Hence it can be supposed the existence of a second reaction generating weight loss in the process.

At the end of the decomposition, the model predicted a final weight percentage of 28.16% in contrast to the 28.18% achieved at the end of thermogravimetric analysis. The prediction itself is extremely accurate and reinforces that the model is able to replicate TGA data at reducing conditions with great fidelity.

Figure 08 demonstrates the effects that the addition of the reducing agent has in the simulations.

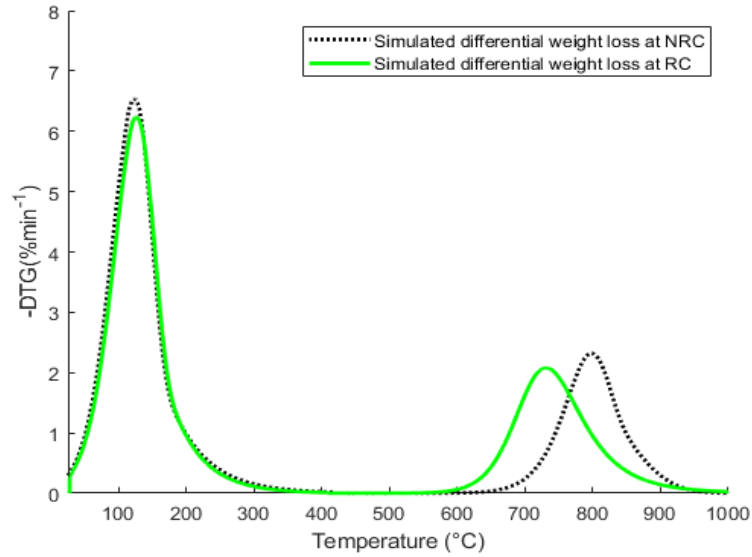


Figure 08 Comparison between simulated DTG curves at reducing and non-reducing conditions

As discussed before, the thermogravimetric analysis in the presence of coke breeze, provided a DTG curve extremely similar to the one obtained in the absence of it, with the main difference being the initial temperature of the decomposition transformation.

Figure 08 demonstrates that the model's simulations were able to capture this behavior. It is clear by analyzing the Figure that the profiles are equal in the dehydration process and only deviate at the decomposition region, due to the effect of the reducing agent. Nevertheless, the profiles observed within this region still resemble a lot, with the major differences observed being the initial temperatures predicted by both curves, and the differential weight loss peaks achieved. It is worthy to mention that, although the predicted beginning temperatures differ, the ending temperatures predicted at reducing and non-reducing conditions, share a high resemblance. This similarity is due to the late

prediction of model's fits at reducing conditions, for the finishing temperature of the decomposition process.

Even if both simulated curves do not succeed in replicating its associated DTG experimental curves, they at least are able to capture and differentiate the effect that the presence of absence of the reducing agent have in the reaction system. This is important since it demonstrates consistency within the models' simulations.

The kinetic parameters of the mathematical model utilized concerning reducing conditions, were estimated by the adjustment of it with the experimental TGA data obtained in the presence of coke breeze. It is important to reassure that the Equations 01 and 02 were utilized for both reducing and non-reducing conditions in order to describe the thermal decomposition's weight loss. Since we readjusted the model for the TGA data considering reducing conditions, the kinetic parameters regarding Equation 01 and 2 were recalculated. To differentiate the parameters calculated for Equation 01 and 02 at reducing and non-reducing conditions, these equations at reducing conditions will be referred as Equation 01' and Equation 02'. Table 05 displays the values of the kinetic parameters regarding each proposed reaction by the model at reducing conditions. The units of k_0 for both Equation 01' and Equation 02' are s^{-1} . The unit of this parameter in Equation 05 is $mol^{-1.5}.s^{-1}$.

Table 05. Estimation of the model's unknown parameters at reducing conditions.

Equations	k_0	$E_a(kJ.mol^{-1})$
1'	$1.637 \cdot 10^5$	41.75
2'	0.214	1.421
5	$5.989 \cdot 10^8$	270.2

For the dehydration process, the obtained parameters for Equation 01' and Equation 02' were practically identical to the ones obtained at non-reducing conditions.

As mentioned in previous topics, at the dehydration region, the experimental TGA curves for both reducing and non-reducing conditions have only minor differences, since the reducing agent does not affect dehydration. Since the same equations were considered to describe the weight loss at this region, it was more than expected that the obtained parameters would be the same for both conditions. Thence, all that was discussed for non-reducing conditions about the results obtained for these parameters also apply in this topic. These obtained values at least show that the parameters estimation process was consistent in its results

Equation 05 describes the decomposition of anhydrous potassium alum. Its main considerations are that aluminum sulfate is not produced, and that carbon participates in the reaction, being consumed to originate carbon monoxide. The estimated Arrhenius constant and activation energy for the reaction regarding this equation were of $5.989 \cdot 10^8 \text{mol}^{-1.5} \cdot \text{s}^{-1}$ and $270.2 \text{kJ} \cdot \text{mol}^{-1}$.

These values sign for interesting conclusions regarding what is depicted in Figure 07. The value estimated for the Arrhenius constant was inferior to the one calculated for Equation 03, and explains the reason why the model predicted an ending temperature above 900°C for the decomposition transformation. At non-reducing conditions, it was attested that the decomposition of the anhydrous potassium alum, represented by Equation 03, terminated at 830°C . Since the calculated k_0 of Equation 05 is much smaller than the one obtained for Equation 03, it is reasonable for the model's prediction of the ending temperature of the decomposition at reducing conditions to be later than 830°C . Even so, this late prediction does not agree with experimental data, which defines 850°C as the ending temperature of the decomposition.

The estimated activation energy had also an interesting value. As the decomposition at reducing conditions begins at a lower temperature than at non-reducing

conditions, which is well demonstrated by Figure 08, it was awaited that the activation energy value of Equation 05 would also present an inferior value when compared to at least the one estimated for Equation 03. The parameters estimation confirmed the generated expectations. The value estimated for the activation energy of Equation 03 exceeds in $7\text{kJ}\cdot\text{mol}^{-1}$ the activation energy calculated for Equation 05. This decay in the decomposition activation energy provoked by the addition of the reducing agent is the reason why, at reducing conditions, this transformation begins sooner. With a diminished E_a , less energy is required to ignite the reaction, and thus, the reaction can be performed at lower temperatures. It is important to remark as well that, the activation energy observed for the transformation of aluminum sulfate into aluminum oxide is still lower than the one estimated for Equation 05, but since this reaction can only start in the presence of aluminum sulfate, it only begins alongside Equation 03, and ,consequently, at a higher temperature. Hence, this noticed discrepancy between the estimated activation energy values and the beginning temperatures are not due to model's flaws.

As well as it was done for non-reducing conditions, the estimated values of the kinetic parameters were used in order to simulate the composition of each component in the reaction medium with the temperature raise. The input composition considered was of 1kmol of potassium alum dodecahydrate and 1.5kmol of carbon, which means that the input composition was stoichiometric. Figure 09 depicts the total solids composition predicted by the model within the advance of temperature in the thermal decomposition.

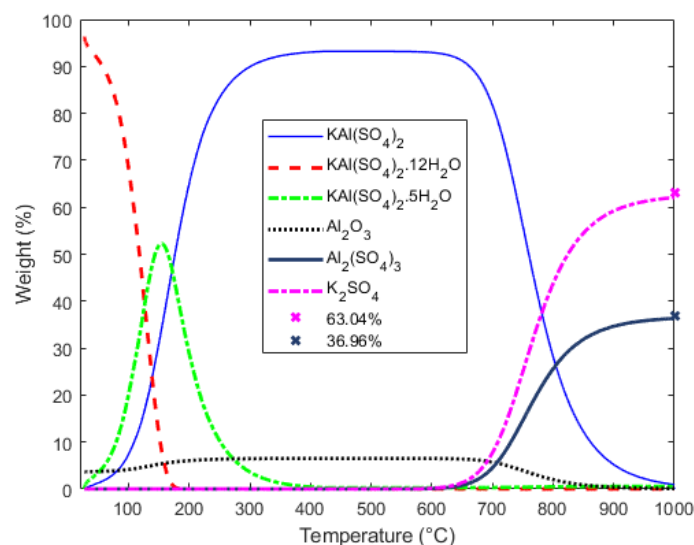


Figure 09 Model's predicted solids composition at reducing conditions

It can be seen that at the dehydration region, the solids composition behavior is the same as observed at non-reducing conditions. The observed composition is slightly different due to the presence of carbon, which remains with stagnant composition during this transformation. At the decomposition region, it was noticed that the formation of the products of interest begins next to 600°C as a result to the effect of carbon, and both potassium sulfate and aluminum oxide are originated following a similar profile due to the absence of aluminum sulfate in the model's prediction.

What becomes very evident in Figure 09, is the model's lack of precision in the prediction regarding the ending temperature of the decomposition transformation. TGA data lead to the conclusion that the thermal decomposition of potassium alum dodecahydrate would be concluded at 850°C. By glancing at Figure 09, it is noticed that the model's prediction deviate a lot from this temperature, as a considerable percentage of potassium alum is still detected at temperatures beyond 900°C

At 850°C, approximately 12% of anhydrous potassium alum are accounted in the total solid's composition, which is equivalent to 70mols of this compound. Even so,

considering that the initial input was of 1kmol of potassium alum dodecahydrate, a conversion of more than 90% is depicted at this temperature. It is important to highlight then, that the high contribution that this compound has in the total solid's composition percentage is due to its superior molar mass when compared to the other two solids present in the system at the end of the decomposition. More information in the model's predicted compositions with the advance of temperature are displayed in Appendix I.

At the end of the simulated thermal decomposition, the total solids composition in the reaction medium consisted in 36.62% of aluminum oxide and 62.69% of potassium sulfate. These results are an outstanding prediction of the system's final composition since as shown before, literature works proposed that at the end of the decomposition, a final composition 63.06% and 36.96% of potassium sulfate and aluminum oxide respectively, would be encountered in the reaction medium.

4. Conclusions

The estimated kinetic parameters were mostly in agreement with previous thermodynamics investigations regarding the thermal decomposition of potassium alum dodecahydrate and the developed model was capable of simulate TGA and DTG data very similar to the ones measured experimentally with great precision.

The mathematical models' fits with the estimated parameters for both reducing and non-reducing conditions presented weight percentage results shared high resemblance to the experimental thermogravimetric analysis data, presenting very minor deviations. The biggest difference observed between a model's predicted weight percentage and a measured value of thermogravimetric analysis was of 3.76 % considering both reducing and non-reducing conditions, which is a great result. Hence, the model and its estimated parameters can be used to perform simulations for this reaction system in the presence or absence of the reducing agent.

The model's final prediction for weight percentages was of 29.11 % and 28.16 %, for non-reducing and reducing conditions, which agreed with the final measurements of the thermogravimetric experiments for both conditions. The model also managed to predict with high fidelity the beginning temperatures of the decomposition of the anhydrous potassium alum for both conditions, although it failed in its predictions regarding the dehydration process. One of the major accomplishments achieved by the model was its final predictions of the total solids composition of the system. The model predictions for both reducing and non-reducing conditions reached a final composition that strongly resonated with the final solids' composition proposed by literature. The behavior of all the species participant in the reaction system was also well depicted by the model's simulations.

As a form of improving the work performed in this paper and help adding more data in the researches regarding the thermal decomposition of potassium alum dodecahydrate, the development of a new and more sophisticated mathematical model to describe this process can be suggested, especially at reducing conditions. The analysis of the experimental DTG curve in the presence of the reducing agent, led to the conclusion that perhaps this transformation may occur in multiple stages, whereas the model utilized proposed a single stage transformation.

Finally, the thermodynamics assessment results show the importance of the addition of the reducing agent in order to accomplish the thermal decomposition of potassium alum. This compound has a drastic effect in the equilibrium compositions of the solid species participant in this process, and strongly decreases the temperature need in order to fulfill a conversion into the products of interest. The reducing agent is a key component in the search for an optimal condition to conduct the glauconite route processing, and to produce efficient fertilizers at competitive market prices.

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Appendix I

Model's predictions at non-reducing conditions

In this topic, the model's prediction for the composition of each compound participant in the thermal decomposition within the advance of temperature is demonstrated and discussed. To calculate each composition, an input of 1kmol of $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ was considered.

Figures A.1 provides a picture into composition of the species participant in the dehydration transformation

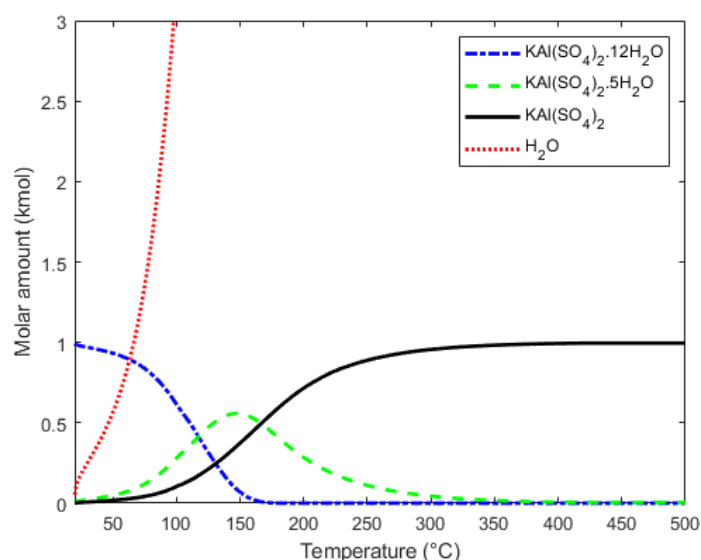


Figure A.1 Model's predicted compositions at the dehydration

It is noticeable in Figure A.1 the lack of information regarding the molar amount of other components such as potassium sulfate, aluminum sulfate, aluminum oxide, SO_2 and O_2 . Since the model considers that these components do not participate in the dehydration, they were not accounted in the elaboration of Figure A.1. Their molar amounts will be depicted further into this work, when the model's predictions for the

decomposition transformation will be discussed. For now, the molar amount of these compounds that are not displayed in Figure A.1 is considered to be equal to zero.

Figure A.1 highlights many things that were already discussed in previous topics. As depicted and discussed before, the consumption of the hydrated alum commences at temperatures close to 20°C. This Figure confirms what was inferred about the beginning of the second dehydration step proposed by the model, as it is observed that the production of potassium alum starts simultaneously with the production of the partially hydrated potassium alum.

With the increase in the temperature values, it is noticed that potassium alum dodecahydrate is continuously converted into $\text{KAl}(\text{SO}_4)_2 \cdot 5\text{H}_2\text{O}$ and water. This component reaches a molar amount peak at the temperature of 150°C. Since $\text{KAl}(\text{SO}_4)_2 \cdot 5\text{H}_2\text{O}$ shares the same stoichiometry of $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, it was expected that at this peak, a value of 1kmol would be observed for the partially hydrated alum. However, the simultaneous conversion of this same compound into potassium alum, makes this value not be achieved by $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$.

At a temperature a little over 150°C $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ is fully consumed and the observed compositions of $\text{KAl}(\text{SO}_4)_2 \cdot 5\text{H}_2\text{O}$ and $\text{KAl}(\text{SO}_4)_2$ are equal. From this point on, the molar amount of $\text{KAl}(\text{SO}_4)_2 \cdot 5\text{H}_2\text{O}$ keep on decreasing while the molar number of $\text{KAl}(\text{SO}_4)_2$ keep on increasing, as the former is being converted in the latter. The dehydration finishes at a temperature a little above 300°C, as seen before, when the molar composition of $\text{KAl}(\text{SO}_4)_2 \cdot 5\text{H}_2\text{O}$ reaches next to zero and stops decreasing.

At Figure A.1 it was chosen not to fully depict the molar amounts of water within the temperature advance with the intent to construct a more concise graph. Yet, since this component is expected to be produced in the two reactions proposed by the model, its

molar amount starts to increase with the consumption of $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, and only ceases to raise at temperature of 300°C , when the dehydration ends.

At the end of this transformation, an amount of approximately 995.4mol of $\text{KAl}(\text{SO}_4)_2$ was produced, and 11.99kmol of water were evacuated of the system, generating the weight loss observed.

The decomposition of potassium alum follows the dehydration transformation. Figure A.2 depicts the species composition with the increase in temperature at the decomposition of potassium alum.

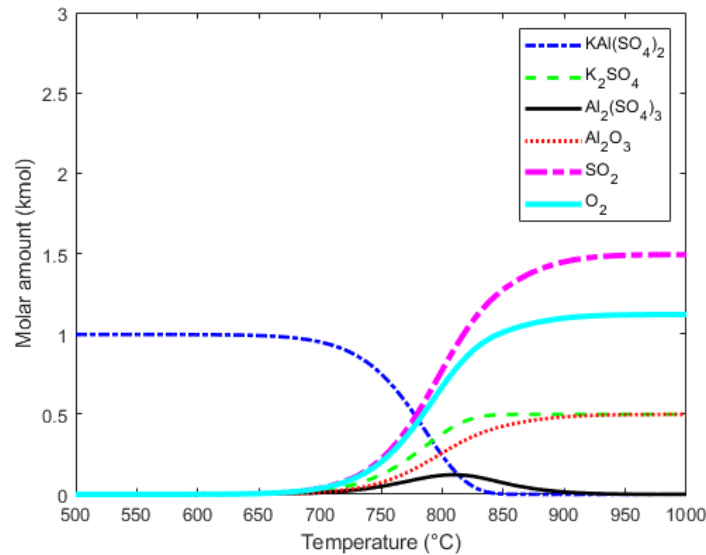


Figure A.2 Model's predicted compositions at the decomposition

By investigating Figure A.2, it is confirmed that the model predicts that $\text{KAl}(\text{SO}_4)_2$ stays stable until the temperature in which the decomposition begins, which is at 680°C . After this temperature, its molar amount starts to decay while the molar amount of the other components increases.

It is interesting to notice that the model's simulated compositions detect only minor formation of aluminum sulfate. As indicated in the in parameters estimation topic,

the obtained activation energy regarding the reaction that contemplated the conversion of this salt into aluminum oxide, was lower than the one estimated for the decomposition of potassium alum, implying that the reactions would occur simultaneously. This becomes very clear when analyzing Figure A.2, as both the molar amounts of aluminum sulfate and oxide start to increase at the same temperatures. This simultaneous production and consumption of aluminum sulfate result in a reduced formation of this compound. When Equation 03 is analyzed, and an initial input of 1kmol of potassium alum is considered, it is deduced that, at a full conversion scenario, 0.250kmol of this salt should be originated in the decomposition reaction. Nevertheless, since the model's prediction signalize that Equation 03 and 04 occur simultaneously, the amount of aluminum sulfate in the model's predictions do not reach 0.25kmol, only achieving a maximum value of 0.122kmol at 809°C.

After the beginning of the decomposition, potassium alum continues to decrease until its composition reaches zero nearly at 830°C. This temperature does not mark the end of the decomposition though. As seen in the previous topic, the estimated values of the Arrhenius constants of Equation 03 and Equation 04 were highly discrepant, with the former being much higher than the latter. This is well observed in Figure A.2. While the decomposition of anhydrous potassium alum terminates at 830°C, the full consumption of aluminum sulfate only terminates at the range of 900-920°C. This is due to the differences in the estimated Arrhenius constants. Since the estimated constant for Equation 03 is much higher, and both reaction commence at the same time, it is reasonable that Equation 03 would finish before Equation 04.

The generated products of this process, with the exception of aluminum sulfate (that is consumed in Equation 04) and potassium sulfate(which does not participate in Equation 04), all share a similar profile regarding its molar amounts. This observed profile

the opposite of the one observed for potassium alum with the main difference observed lying within the final molar amounts achieved by each component.

At the end of the thermal decomposition, the model predicted that the presence aluminum sulfate and potassium alum would not be accounted as both these compounds would have already been consumed. The molar amounts regarded to the gases responsible for the weight loss would be of approximately 1.49 kmol of SO_2 and 1.12 of O_2 . Finally, the molar amount shared by the products of interest would be of approximately 0.497kmol.

Model's predictions at reducing conditions

At this final section, the model's predicted compositions at reducing conditions with the advance in temperature is discussed. Since the predicted compositions in the dehydration region at reducing conditions are very similar with the ones at non-reducing conditions, the predictions regarding dehydration will not be approached at this section.

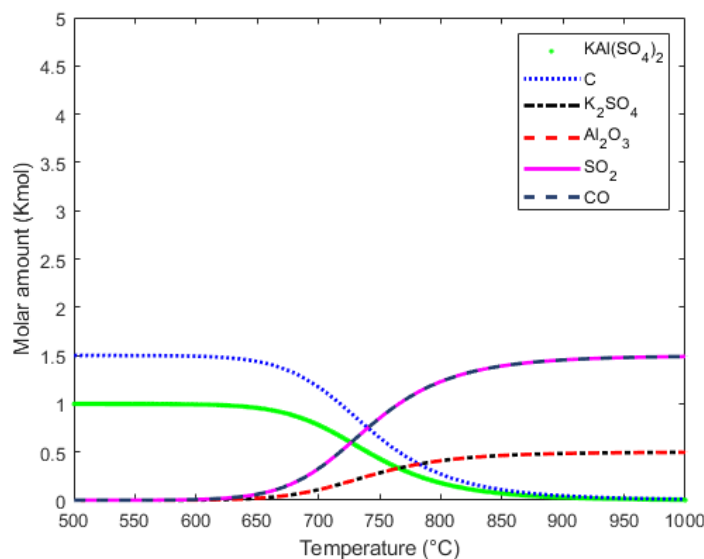


Figure A.3 Model's predicted compositions at the decomposition for reducing conditions

Figure A.3 displays the molar amount of each component participant in the decomposition as a function of temperature. The initial input was of 1kmol of potassium alum dodecahydrate and 1.5kmol of Carbon.

The effect of carbon in the reaction can be attested right away by the temperature in which the formation the consumption of the reagents begins. After the beginning of the reaction, the reagents are consumed until the transformation's end. This Figure bolsters that the actual prediction of the decomposition's end by the model was a little late, since the molar amount of alum reaches zero at a temperature above 900°C.

The absence of aluminum sulfate in the proposed model makes all the profiles regarding the product species in the reaction system to be similar. At the end, approximately 0.499kmol of potassium sulfate and aluminum oxide is formed, with 1.496kmol of SO₂ and carbon monoxide being eliminated from the reaction system and provoking the weight loss in the process. All the reagents in the transformation are almost fully consumed within the transformation, with only very few remnants of them present in the reaction system at the end of the thermal decomposition.